

Cambridge International Examinations
Cambridge International Advanced Subsidiary and Advanced Level

CANDIDATE
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PHYSICS

9702/23

Paper 2 AS Level Structured Questions

October/November 2017

1 hour 15 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer **all** questions.

Electronic calculators may be used.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Data

speed of light in free space	$c = 3.00 \times 10^8 \text{ m s}^{-1}$
permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$
permittivity of free space	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$
	$(\frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ m F}^{-1})$
elementary charge	$e = 1.60 \times 10^{-19} \text{ C}$
the Planck constant	$h = 6.63 \times 10^{-34} \text{ J s}$
unified atomic mass unit	$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$
rest mass of electron	$m_e = 9.11 \times 10^{-31} \text{ kg}$
rest mass of proton	$m_p = 1.67 \times 10^{-27} \text{ kg}$
molar gas constant	$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
the Avogadro constant	$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$
the Boltzmann constant	$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
acceleration of free fall	$g = 9.81 \text{ m s}^{-2}$

Formulae

uniformly accelerated motion

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas

$$W = p\Delta V$$

gravitational potential

$$\phi = -\frac{Gm}{r}$$

hydrostatic pressure

$$p = \rho gh$$

pressure of an ideal gas

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

simple harmonic motion

$$a = -\omega^2 x$$

velocity of particle in s.h.m.

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

Doppler effect

$$f_o = \frac{f_s v}{v \pm v_s}$$

electric potential

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor

$$W = \frac{1}{2} QV$$

electric current

$$I = Anvq$$

resistors in series

$$R = R_1 + R_2 + \dots$$

resistors in parallel

$$1/R = 1/R_1 + 1/R_2 + \dots$$

Hall voltage

$$V_H = \frac{BI}{ntq}$$

alternating current/voltage

$$x = x_0 \sin \omega t$$

radioactive decay

$$x = x_0 \exp(-\lambda t)$$

decay constant

$$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$$

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Answer **all** the questions in the spaces provided.

- 1 (a) (i) Define *power*.

.....
[1]

- (ii) Show that the SI base units of power are $\text{kg m}^2 \text{s}^{-3}$.

[1]

- (b) All bodies radiate energy. The power P radiated by a body is given by

$$P = kAT^4$$

where T is the thermodynamic temperature of the body,
 A is the surface area of the body
 and k is a constant.

- (i) Determine the SI base units of k .

base units[2]

- (ii) On Fig. 1.1, sketch the variation with T^2 of P . The quantity A remains constant.

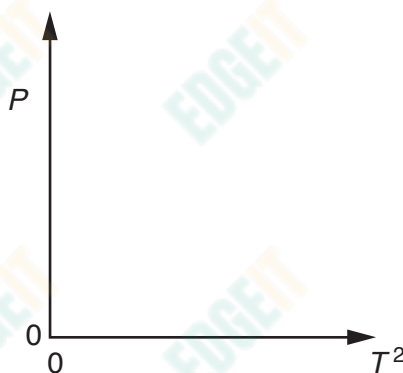


Fig. 1.1

[1]

[Total: 5]

- 2 A liquid of density ρ fills a container to a depth h , as shown in Fig. 2.1.

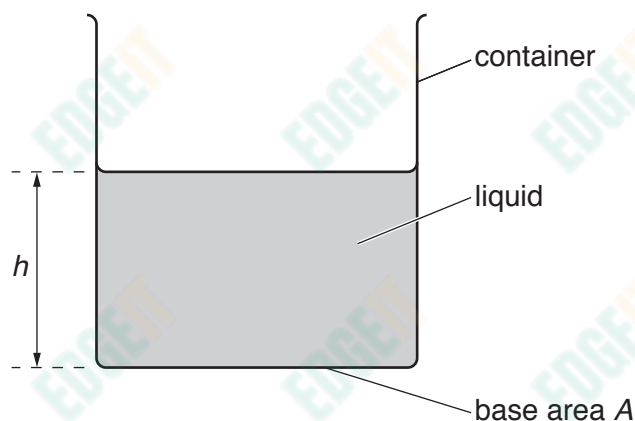


Fig. 2.1

The base of the container has area A .

- (a) Derive, from the definitions of pressure and density, the equation

$$p = \rho gh$$

where p is the pressure exerted by the liquid on the base of the container and g is the acceleration of free fall.

[3]

- (b) A small solid sphere falls with constant velocity through the liquid.

- (i) State

1. the names of the three forces acting on the sphere,

.....

.....

2. a word equation that relates the magnitudes of these forces.

.....

[2]

- (ii) State and explain the changes in energy that occur as the sphere falls.

.....

.....

.....

.....[2]

- (c) The liquid in the container is liquid L. Liquid M is now added to the container. The two liquids do not mix. The total depth of the liquids is 0.17 m.

Fig. 2.2 shows how the pressure p inside the liquids varies with height x above the base of the container.

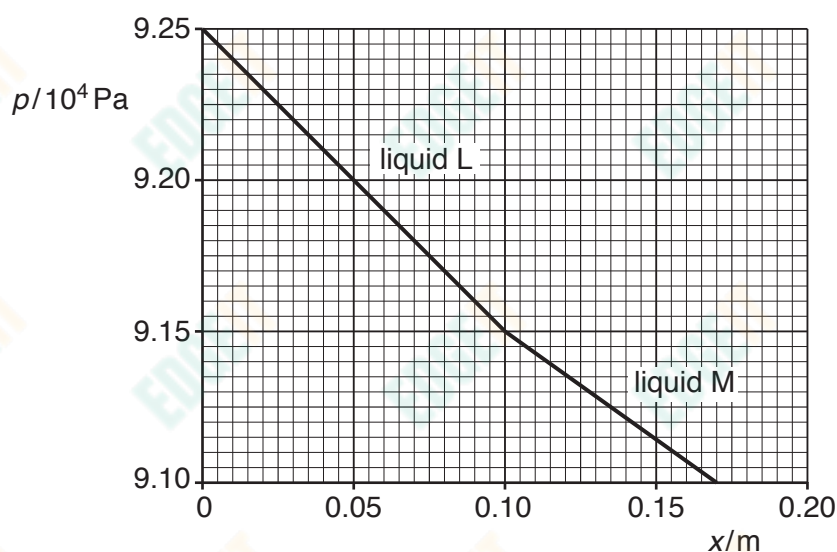


Fig. 2.2

Use Fig. 2.2 to

- (i) state the value of atmospheric pressure,

atmospheric pressure = Pa [1]

- (ii) determine the density of liquid M.

density = kg m^{-3} [2]

[Total: 10]

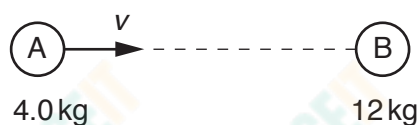
- 3 (a) State the principle of conservation of momentum.

.....

.....

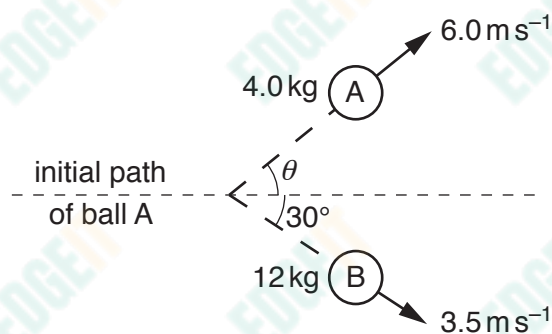
.....[2]

- (b) Ball A moves with speed v along a horizontal frictionless surface towards a stationary ball B, as shown in Fig. 3.1.



before collision

Fig. 3.1



after collision

Fig. 3.2 (not to scale)

Ball A has mass 4.0 kg and ball B has mass 12 kg .

The balls collide and then move apart as shown in Fig. 3.2.

Ball A has velocity 6.0 ms^{-1} at an angle of θ to the direction of its initial path.

Ball B has velocity 3.5 ms^{-1} at an angle of 30° to the direction of the initial path of ball A.

- (i) By considering the components of momentum at right-angles to the direction of the initial path of ball A, calculate θ .

$\theta = \dots\dots\dots^\circ$ [3]

- (ii) Use your answer in (i) to show that the initial speed v of ball A is 12 m s^{-1} .
Explain your working.

[2]

- (iii) By calculation of kinetic energies, state and explain whether the collision is elastic or inelastic.

[3]

[Total: 10]

- 4 (a) By reference to the direction of propagation of energy, explain what is meant by a *longitudinal* wave.

.....
[1]

- (b) A car horn emits a sound wave of frequency 800 Hz. A microphone and a cathode-ray oscilloscope (c.r.o.) are used to analyse the sound wave. The waveform displayed on the c.r.o. screen is shown in Fig. 4.1.

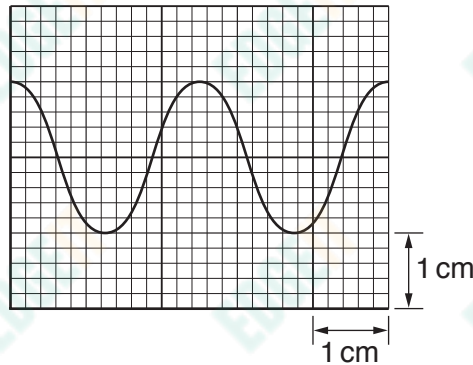


Fig. 4.1

Determine the time-base setting, in s cm^{-1} , of the c.r.o.

time-base setting = s cm^{-1} [3]

- (c) The intensity I of the sound at a distance r from the car horn in (b) is given by the expression

$$I = \frac{k}{r^2}$$

where k is a constant.

Fig. 4.2 shows the car in (b) on a road.

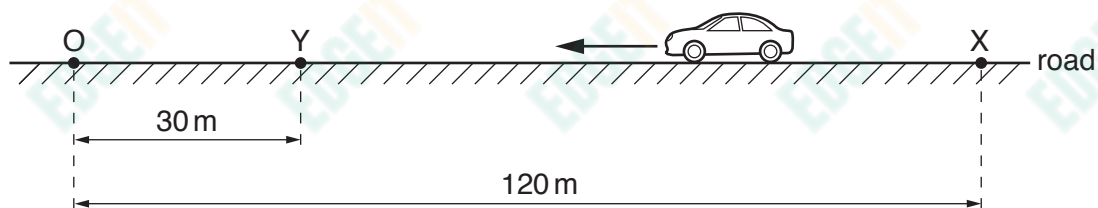


Fig. 4.2

An observer stands at point O. Initially the car is parked at point X which is 120 m away from point O. The car then moves directly towards the observer and stops at point Y, a distance of 30 m away from O.

The car horn continuously emits sound when the car is moving between points X and Y.

- (i) The sound wave at point O has amplitude A_X when the car is at X and has amplitude A_Y when the car is at Y.

Calculate the ratio $\frac{A_Y}{A_X}$.

ratio = [3]

- (ii) When the car is parked at X, the frequency of the sound from the horn that is detected by the observer is 800 Hz. As the car moves from X to Y, the maximum change in the detected frequency is 16 Hz. The speed of the sound in air is 330 m s^{-1} .

Determine, to two significant figures,

1. the minimum wavelength of the sound detected by the observer,

wavelength = m [2]

2. the maximum speed of the car.

speed = m s^{-1} [2]

[Total: 11]

- 5 (a) Define *electric field strength*.

.....
[1]

- (b) Two parallel metal plates in a vacuum are separated by 0.045 m. A potential difference V is applied between the plates, as shown in Fig. 5.1.

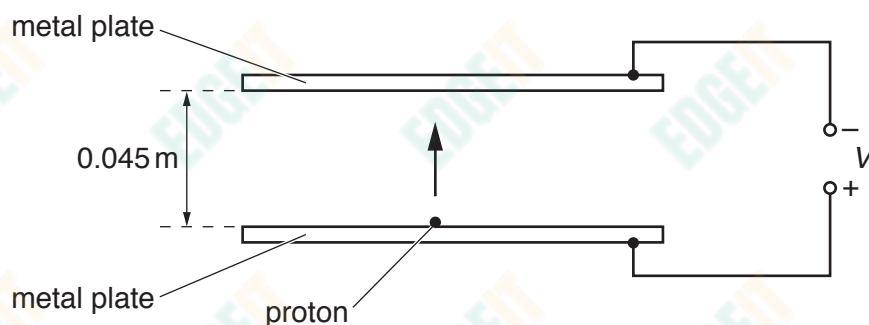


Fig. 5.1

A proton is initially at rest on the surface of the positive plate. The proton in the uniform electric field takes a time of 1.5×10^{-7} s to reach the negative plate.

- (i) Show that the acceleration of the proton is $4.0 \times 10^{12} \text{ m s}^{-2}$.

[2]

- (ii) Calculate the electric force on the proton.

force = N [1]

(iii) Use your answer in (ii) to determine

1. the electric field strength,

field strength = NC^{-1} [2]

2. the potential difference V between the plates.

$V = \dots\dots\dots$ V [2]

(c) An α particle is now accelerated between the two metal plates in (b) by the electric field.

Calculate the ratio

$$\frac{\text{acceleration of } \alpha \text{ particle}}{\text{acceleration of proton}} .$$

ratio = [2]

[Total: 10]

6 A filament lamp is rated as 30 W, 120 V. A potential difference of 120 V is applied across the lamp.

(a) For the filament wire of the lamp, calculate

(i) the current,

current = A [2]

(ii) the number of electrons passing a point in 3.0 hours.

number = [2]

(b) Show that the resistance of the filament wire is $480\ \Omega$.

[2]

(c) The filament wire has an uncoiled length of 580 mm and is made of metal. The metal has resistivity $6.1 \times 10^{-7}\ \Omega\text{ m}$ at the operating temperature of the lamp.

Calculate the diameter of the wire.

diameter = m [3]

(d) The potential difference across the lamp is now reduced. State and explain the effect, if any, on the resistance of the filament wire.

.....
 [1]

[Total: 10]

- 7 (a) A nucleus X decays by emitting a β^+ particle to form a new nucleus, ${}^{23}_{11}\text{Na}$.

State the number of nucleons and the number of neutrons in nucleus X.

number of nucleons =

number of neutrons =

[2]

- (b) State one similarity and one difference between a β^+ particle and a β^- particle.

similarity:

difference:

[2]

[Total: 4]

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